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Influence of Aeration, Introduction of Probiotics, and Supply of Water on Landfill Gas Production—Study of Models

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Abstract: When municipal solid waste (MSW) is placed in a landfill, it undergoes anaerobic decomposition, leading to the production of landfill gas, which primarily consists of methane (CH_4) and carbon dioxide (CO_2). Reducing methane emissions is essential in the fight against climate change. It must be implemented at global and European levels, as set out in 2030 in the impact assessment of the climate goal plan. This assessment states that to achieve the goal by 2030 and to reduce greenhouse gas emissions by at least 55%, the methane emissions must be reduced, considering the goals of the Paris Agreement. The Glasgow Climate Pact includes a global mitigation target of the year 2030: to reduce CO_2 emissions by 45%, and the emissions of methane and other greenhouse gasses. For that purpose, looking for new, more advanced ways of managing such waste is necessary. The main objective of this experimental study was to evaluate the influence of aeration, probiotic introduction, and water supply on the production of landfill gasses (CO_2, CH_4, N_2, H_2, etc.) in five different landfill models during the management of MSW and to propose the best solutions for reducing environmental pollution. The results of the research showed that the first and second models of landfills, using only anaerobic conditions, can be used for the treatment of MSW for the production of biogas (CH_4, CO_2), as up to 40–60% of it was released during the 120-experiment period. The third landfill model can be applied in old, already closed landfills, where the rapid stabilization and aeration of MSW are required to minimize pollutant emissions (N_2, etc.) and unwanted odors and shorten biodegradation processes. The results of the fourth and fifth landfill models, in which aerobic–anaerobic conditions were applied, showed that the developing nitrification–denitrification processes resulted in complete nitrogen removal (from 20% to 0%), and overall waste stabilization improved the biodegradation of the MSW. Later, relatively good (on average, 30%) results of biogas (CH_4, CO_2) emissions are achieved during anaerobic condition formation results. Summarizing all experiment results of all landfill models for the further evaluation of the processes, all models can be applied in real practice depending on where they will be used and what result they want to achieve.

Keywords: municipal solid waste; landfill; greenhouse gas emissions; biodegradable waste; aeration; probiotics; leachate recirculation

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1. Introduction

In landfills, municipal solid waste (MSW) is often decomposed under anaerobic conditions, during which landfill biogas is formed [1,2]. Methane (CH_4) and carbon dioxide (CO_2) dominate in this biogas [1,3,4]. Based on the data of other scientists, it was found that as much as 16% of methane is released into the atmosphere when organic waste rots in landfills. Reducing methane emissions is essential in the fight against climate change. It must be implemented at global and European levels, as set out in 2030 in the impact assessment of the climate goal plan. This assessment states that to achieve the goal by 2030 and to reduce greenhouse gas emissions by at least 55%, the methane emissions must be reduced, considering the goals of the Paris Agreement. The Glasgow Climate Pact includes a global mitigation target for 2030: to reduce CO_2 emissions by 45% and the emissions of

methane and other greenhouse gasses. In addition to those mentioned (CH_4 and CO_2), different greenhouse gasses (up to 2–10%) are released from landfills. In addition to the above-mentioned chemical compounds, these gasses contain nitrogen, oxygen, ammonia, sulfides, hydrogen, and others [5].

Eurostat data show that Lithuania's annual MSW generation shows a rather stable value of around 1.3 million tonnes since 2016 and 1.35 million tonnes in 2020. Waste generation per capita slowly increased from 444 kg/cap in 2016 to 483 kg/cap in 2020 but remains under the (estimated) EU average of 505 kg/cap. Lithuania reduced its reliance on landfilling by heavily increasing the mixed municipal waste treatment capacity, including mechanical biological treatment (MBT) and incineration (with energy recovery) [6]. For that purpose, looking for new, more advanced ways of managing such waste is necessary. A truly sustainable landfill is one in which waste materials are safely assimilated into the surrounding environment, whether or not they have been treated by biological, thermal, or other processes, and which manages gas-related problems to minimize the environmental impact. In addition, such a landfill must meet all environmental requirements. Based on literature sources [7], the design of an engineering landfill is considered a technical measure, and management measures such as the separation of organic and inorganic waste can be seen as an effective waste management strategy. There are many publications on sustainable landfills about innovative methods to reduce energy consumption, implementing horizontal landfill aeration, or applying the zero-waste concept [8–10].

Municipal solid waste (MSW) landfills are usually located and operated by conventional landfill disposal methods, where uncontrolled anaerobic conditions typically occur. Such uncontrolled waste management can harm the environment and human health, and landfill leachate can seep into groundwater. As a result, after evaluating the disadvantages of traditional waste management practices, detailed laboratory studies are carried out aimed at waste stabilization and filtrate recirculation to reduce waste time, improve filtrate quality, and increase the gas production rate. In the world, there is an increasing focus on enabling new perspectives on waste disposal, which would allow a quick stabilization or detoxification of waste and reduce or increase the amount of methane gas, volatile organic substances, etc. [8–10].

Interesting facts can be found when analyzing the models of sustainable landfills. That is, how to change landfill processes into useful ones and put them into practice. For example, during the operation of landfills, they will produce biogas under anaerobic conditions, which can be used as a renewable energy source. Such a sustainable landfill would provide a practical gas extraction idea to improve the energy extraction needs of the surrounding residential areas, which could give the citizens economic benefits from landfills.

In addition, if the rapid biodegradation of organic waste is required, then it is necessary to maintain aerobic conditions in landfills. During the latter process, there is an opportunity to make compost quickly. Compost produced in this way saves money by ensuring that the operation is sustainable and environmentally friendly. Such aerobic pretreatment of MSW reduces waste mass and improves environmental processes in landfills. Other measures in waste management processes, such as probiotic injection, filtrate recirculation, and choosing an effective biocover, can be attributed to a sustainably managed landfill. For example, filtrate recirculation ensures better nutrient distribution and adequate moisture content. Biocover is one of the aspects of innovative technologies for reducing methane emissions from landfills and achieving landfill sustainability through remediation. The studies of other researchers [8–10] confirmed that a higher amount of leachate and a dose of probiotics can maintain a higher and more stable methane concentration. In summary, a sustainable landfill is one in which all physicochemical and biochemical processes are harmonious. Therefore, the operation of such a landfill is not harmful or utterly harmless to the environment [11–18].

The main objective and novelty of this experimental complex study involved evaluating the influence of aeration, probiotic introduction, and water supply on the production of landfill biogases (CO_2, CH_4, etc.) in the different landfill models, compare their operational

parameters, and propose the best solutions for reducing environmental pollution and creating more sustainable disposal of municipal solid waste.

2. Materials and Methods

The main challenge in mechanical–biological treatment (MBT) municipal solid waste (MSW) research is to predict the total production of biogas (methane, carbon dioxide, etc.) and their cleaning from pollutants so that the operation of different landfills does not harm the environment. After landfilling, MSW goes through degradation and is affected by bacteria. Generally, this works under anaerobic conditions, but overlying layers hold the airflow as more MSW is added, and the dominant biochemical reactions become anaerobic. This study analyzes the behavior of different landfill model implementations for various practical situations. The in situ treatment processes include anaerobic, aerobic, and aerobic–anaerobic bioreactor technology.

MSW for experimental studies was taken from the Kazokiškių municipal waste landfill of Vilnius County, Lithuania. Waterproof and sealed plastic bags were used for each sample storage and transfer to the laboratory. The samples were separated by type and crushed into average 10–20 mm particle sizes. MSW was homogenized before filling the five plastic bottles (with a capacity of 19 L), which were used as five landfill models under different conditions during this research. Each of them was filled with MSW (5 kg), which consisted of paper/cardboard (9%), wood waste (3%), textile waste (6%), food waste (21%), green waste (19%), plastic packaging (12%), metal packaging (2.8%), glass packaging (6.8%), inert waste (10.2%), other non-hazardous MSW (8%), and composite packaging (2.2%). Before being placed in the experimental bottles, all wastes were mechanically treated (i.e., crushed and mixed) as they were typically received in landfills. Waste was held in experimental bottles for 120 days. The technological scheme of the landfill models (anaerobic; anaerobic with a dose of probiotics; aerobic; aerobic–anaerobic; aerobic–anaerobic with a dose of probiotics) is shown in Figure 1, and in Tables 1 and 2. The five different landfill models had different conditions and experimental parameters. Each model was performed in three repetitions.

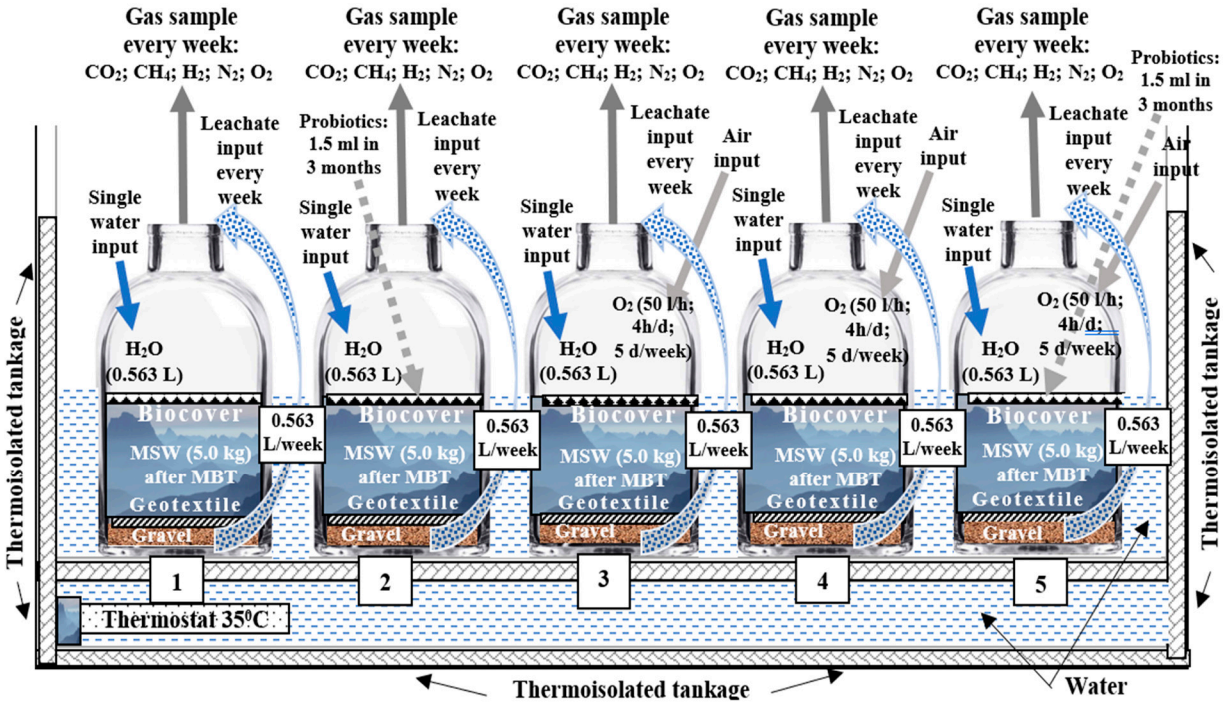


Figure 1. The five different landfill models' technological inputs and output schemes: 1—the anaerobic landfill; 2—the anaerobic landfill with a dose of probiotics; 3—the aerobic bioreactor landfill; 4—the aerobic–anaerobic landfill; 5—the aerobic–anaerobic landfill with a dose of probiotics.

Table 1. The operating conditions of the five different landfill models.

Landfill Models	Material	Operating Conditions	Conditions in the Different Landfill Models
1	MSW after MBT	Anaerobic with a single input of water and leachate recirculation	Anaerobic landfill
2	MSW after MBT	Anaerobic with a single input of water, leachate recirculation, and dose of probiotics	Anaerobic landfill
3	MSW after MBT	Aerobic with high air inflow, single input of water, and leachate recirculation	Aerated bioreactor landfill
4	MSW after MBT	Aerobic–anaerobic with low air inflow, single input of water, and leachate recirculation	Aerobic–anaerobic landfill
5	MSW after MBT	Aerobic–anaerobic with low air and single input of water, leachate recirculation, and dose of probiotics	Aerobic–anaerobic landfill

Table 2. Experimental parameters of the five different landfill models during this study.

Landfill Models	MSW after MBT (kg)	Air Inflow	First Added Water Flow (L)	Leachate Input (L/week)	Probiotics (Liquid) (L)
1	5.0	–	0.563	0.563	–
2	5.0	–	0.563	0.563	One dose of 1.5 mL
3	5.0	50 L/h, 4 h/d, 5 d/week	0.563	0.563	–
4	5.0	50 L/h, 4 h/d, 5 d/week	0.563	0.563	–
5	5.0	50 L/h, 4 h/d, 5 d/week	0.563	0.563	One dose of 1.5 mL

All bottoms of each bottle were filled with gravel (suitable for drainage), on which a layer of a geotextile was placed. The prepared waste was composed in the bottles on the mentioned layers. Later, it was covered by a one-year-old, high-quality, stable layer of compost (called biocover). This biocover for all models consisted of biodegradable waste such as tree branches, grass, etc. These layers correspond to the structures of the actual landfill.

These bottles were closed and installed into a thermoisolated tankage filled with water. The water was kept at a temperature of 35 °C and measured by a thermostat. The third, fourth, and fifth landfill models that required aeration were additionally equipped with aerators.

The gasses generated inside five landfill models were collected weekly into 10 L—"Tedlar" (type 232 SKC, Emmett, MI, USA)—gas sampler bags. Measurements of biogas composition were made using the biogas analyzer INCA 4000 (UNION Instruments GmbH, Karlsruhe, Germany), which presented methane (CH_4, %), carbon dioxide (CO_2, %), and oxygen (O_2, %) concentrations. Determined concentrations were within the following measure range: oxygen—0–25% (error—±1%), methane—0–100% (error—±1%), and carbon dioxide—0–100% (error—±1%). The nitrogen (N_2, %) and hydrogen (H_2, %) concentrations were determined using a portable biogas analyzer model, SKZ1050C-CH_4/H_2/N_2 (SKZ International Co., Ltd., Jinan, China). Determined concentrations were within the following measure range: nitrogen—0–100% (error—±1%), hydrogen—0–1000 ppm (error—±1). The pH was determined using the pH meter, with an error of ±0.01. The pH was determined using the Mettler Multi Seven pH meter (Mettler-Toledo GmbH, Giessen, Germany), with an error of ±0.01. The required value was determined by a potentiometric method (according to the standard ISO 10390:2005) [19].

Probiotics were used in experimental studies. The probiotics were produced through natural fermentation but not chemically synthesized or genetically engineered. These probiotics in the liquid phase were created through natural fermentation using beneficial and effective microorganisms. The general benefits of probiotic injection were

- To increase the productivity of the waste stabilization process;

- To improve air quality by increasing the biodegradability and fermentation of the waste;
- To achieve reductions in odors from landfill models.

Water (0.563 L) was supplied to each landfill model during the experiments. The collected air emissions were analyzed every week. At the same time, the collected filtrate was returned to each landfill model and reused in its irrigation process. Anaerobic conditions were maintained in the first and second landfill models. The second and fifth landfill models were filled with 1.5 mL of "Odor Away" probiotics. Aerobic conditions were left in the third landfill model. The fifth landfill model maintained aerobic–anaerobic conditions. The third, fourth, and fifth landfill models were supplied with 50 L of O_2 (four hours per day and five days per week). This landfill models have an air compressor called "Oxyboost 100" (AQUAEL Sp. z o. o., Suwalki, Poland). All experimental parameters of the five different landfill models during the study are given in Table 2.

The leachate was recirculated once a week. In total, 0.563 L of clean water (Table 2) was added to all landfill models filled with leachate to produce leachate. This study used leaching as a closed recirculation system in all landfill models to increase the waste's biodegradability. All landfill models were equipped with four ports. One port was used for drainage. The other two ports were used to collect gas samples and inject liquid and air. The fourth port was used for single water input.

Total organic carbon was determined using the "Shimadzu" carbon analyzer TOC-VCSH (Shimadzu, Corp., Tokyo, Japan). The biological oxygen demand (over five days) (BOD_5) and the chemical oxygen consumption (COD) analysis were carried out according to the methods described in the Lithuanian Standards of Wastewater Examination, which are compatible with the American Public Health Association (APHA) [20,21]. The chemical oxygen consumption (COD) was measured using the fixed titrimetric method. The index of bichromatic oxidation was applied using the thermoreactor Eco-6 of Velp Scientifica (Usmate Velate, Italy).

The average experimental results were calculated from three replicates of each experimental treatment and reported as the mean ± standard deviation. The data were analyzed using a variance analysis, and only values with a p-value less than 0.05 were considered acceptable. A variance analysis was performed to determine how the concentrations of the analyzed gas emissions from different landfill models changed over 120 experimental days.

3. Results and Discussion

3.1. The First Landfill Model

When the MSW was placed in the first landfill model, it began to decompose under anaerobic conditions, resulting in the formation of landfill gas. Like all gasses, these gasses are formed due to physical, chemical, and microbial processes occurring in MSW. It was found that in the acetogenesis phase of landfill gas production, due to the activity of anaerobic bacteria, CO_2 (from 25 to 64%) and H_2 (from 13 to 20%) increased rapidly on the 14–35th day of the experiment (Figure 2). Meanwhile, O_2 (from 3 to 0%) and N_2 (from 39 to 18%) decreased significantly, respectively. Later, on days 35–56 of the experiment, the methanogenesis phase began without oxygen in the medium, in which methane-producing methanogenic bacteria started to dominate. In the first landfill model studied, methane emissions increased on day 35 of the experiment. On that basis, at this stage, due to the activity of mesophilic bacteria in the decomposition of MSW, CH_4 production increased from 0 to 53%. However, the formation of all other landfill gasses began to decline, respectively, CO_2—from 64 to 45%, H_2—from 18 to 6%, and N_2—from 20 to 4%. This could have been influenced by the microorganisms living here because methanogens are susceptible microorganisms that can consume acetates, hydrogen, and CO_2 to produce methane. At the end of the methanogenesis process, on the 56th day of the experiment, equal emissions of greenhouse gasses (CH_4 and CO_2) were recorded, which were 45%.

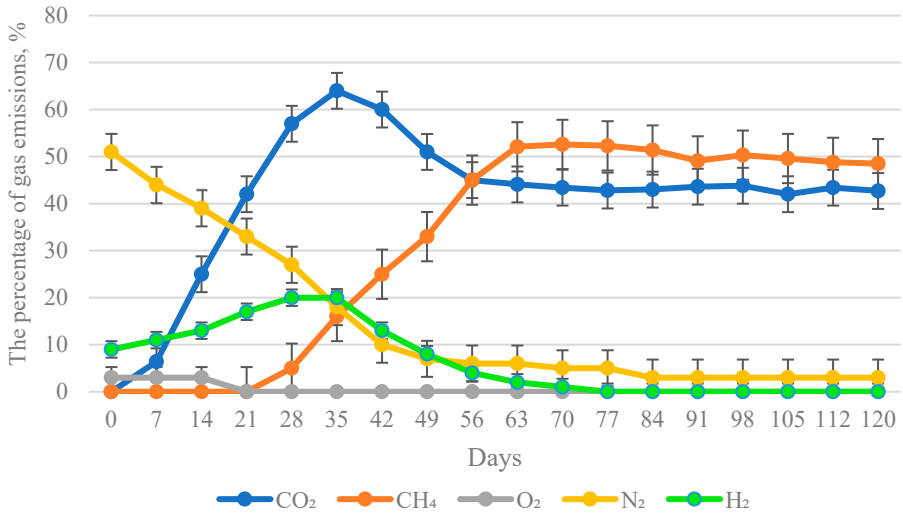


Figure 2. Gas emissions from the first landfill model.

In the anaerobic process, less energy is produced because glucose is not entirely broken down without oxygen. During the anaerobic process, less energy is produced because glucose is not entirely broken down without oxygen. For the methanogenesis stage (from day 56 to day 120 of this study), the emissions of landfill gasses (CH_4 and CO_2) remained relatively stable. Specifically, CH4 emissions fluctuated between 45% and 53%, while CO_2 emissions ranged from 42% to 45%. At the same time, the concentrations of other pollutants decrease. For example, N_2 emissions decreased from 6% to 3%, and H_2 emissions decreased from 4% to 1%.

An increase in pH from 6.6 to 7.2 is observed at the end of the methanogenesis process. In this process, the amount of organic compounds decreased. This was shown by chemical oxygen demand (COD), total organic carbon (TOC), and biological oxygen demand (BOD_5) studies. Accordingly, COD (from 5.00 to 3.00 mg/L), TOC (from 0.60 to 1.00 mg/L), and BOD_5 (from 3.00 to 1.00 mg/L) decreased. These studies were confirmed by the results of other researchers [22–26]. Nag et al. [26] investigated the effect of leachate recirculation on the hydrolysis of food waste using leached bed reactors. They hypothesized that increased recirculation increased the ability of hydrolytic microorganisms to contact solid surfaces.

The anaerobic phase can be used as a renewable energy source, as biogas, consisting of methane, carbon dioxide, and other traces of "polluting" gasses, is produced [22–26]. Researchers Fogarassy et al. [27] found that waste incineration in metropolitan waste treatment facilities is inconsistent with sustainable development. The solution to this problem can appear in the future as waste composting, that is, as a sustainable way of processing it. In addition, compared to traditional landfills, the leachate pollution decreased as organic matter decreased. However, it should be remembered and appreciated that sometimes anaerobic processes involve more chemical steps and take place much more slowly than aerobic ones. During our research, no increased ammonia concentration was recorded in the filtrate. An increase in ammonia concentration can cause a partial or complete suppression of methane production and unpleasant odors, which does not fully meet the requirements of a sustainable and non-polluting landfill.

3.2. The Second Landfill Model

The second anaerobic landfill had different parameters compared to the first one. Probiotics were injected and doubled the amount of injected leachate, which resulted in a 63% increase in methane emissions over the 70 days of the experiment (Figure 3). By the end of the experiment, methane emissions had reached as high as 60–63%. It can be argued that the injection of probiotics resulted in a more even distribution of methane emissions.

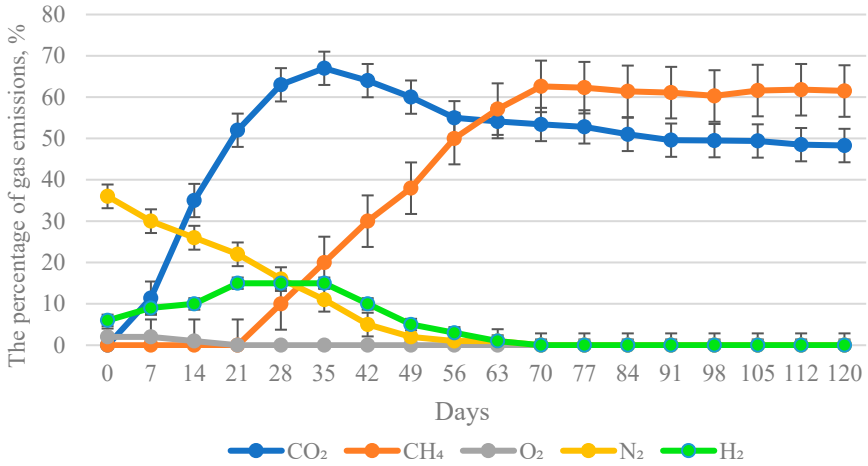


Figure 3. Gas emissions from the second landfill model.

The experiment showed shallow oxygen content (1–2%) during the first 14 days. Afterward, the concentration of this pollutant decreased to zero for the rest of the experiment. As a result, the environment in the landfill model became anaerobic, completely stopping the activity of aerobic microorganisms [23,27–37]. During the initial 35 days of the experiment, the population of anaerobic microorganisms led to a rapid increase in carbon dioxide levels, with emissions ranging from 11% to 67%. Subsequently, like in the first landfill model, the activity of active methanogenic bacteria resulted in a decline in CO_2 emissions from 48% to 67%. Like other researchers [19,22,27–31], this study showed a similar trend with other pollutants. Optimizing anaerobic processes and managing leachate and biogas generated in landfills are aspects of creating more sustainable disposal of municipal solid waste.

For instance, N_2 emissions decreased from 36% at the beginning of the experiment to 1% by the 63rd day. In the acetogenesis phase (between the 1st and 35th day of the experiment), the activity of anaerobic bacteria caused a rapid increase in H_2 concentrations from 6% to 15%. However, these concentrations gradually decreased to a minimum during the methanogenesis stage. Pollutant emissions in the first and second landfill models were similar but different. As previously mentioned, due to changed conditions (additionally injecting probiotics, doubling the amount of filtrate), in the second landfill model, biogas (CO_2 and CH_4) emissions averaged 10–15%, and the duration of the acetogenesis and methanogenesis phases increased to 7–10 days. In the second landfill model, N_2, H_2, and O_2 emissions decreased by 15–20% compared to the first. The pH increased from 5.6 to 6.2, and the COD, TOC, and BOD_5 studies showed similar changes. With the decrease in organic compounds, COD decreased from 3.00 to 2.00 mg/L, TOC decreased from 0.60 to 0.35 mg/L, and BOD_5 decreased from 2.00 to 0.50 mg/L.

3.3. The Third Landfill Model

When the MSW waste was placed in the third landfill model, it was decomposed under aerobic conditions. These conditions were ensured with continuous air injections and a constant recirculation of the filtrate. Nitrogen and hydrogen removal was significantly faster under aerobic conditions than in the first and second landfill models. For example, during the first 70 days of this study, N_2 emissions decreased from 25% to 0% (Figure 4). Aerobic conditions throughout the waste mass quickly reduced methane levels to a minimum. These studies found that methane formation was stopped during these studies, but CO_2 was mainly formed (from 9 to 30%) during the entire study period. For example, the exhaust gas composition was the following: 6–18% O_2, 9–30% CO_2, 3–25% N_2, and 1–5% H_2. Compared to the first landfill models examined, CO_2 was reduced by 10%. Aeration processes significantly reduced COD (from 2.50 to 1.50 mg/L), TOC (from 0.35 to 0.20 mg/L), and BOD_5 (from 1.50 to 0.50 mg/L) of the filtrate. During this process, the pH prevailed between 6 and 8.

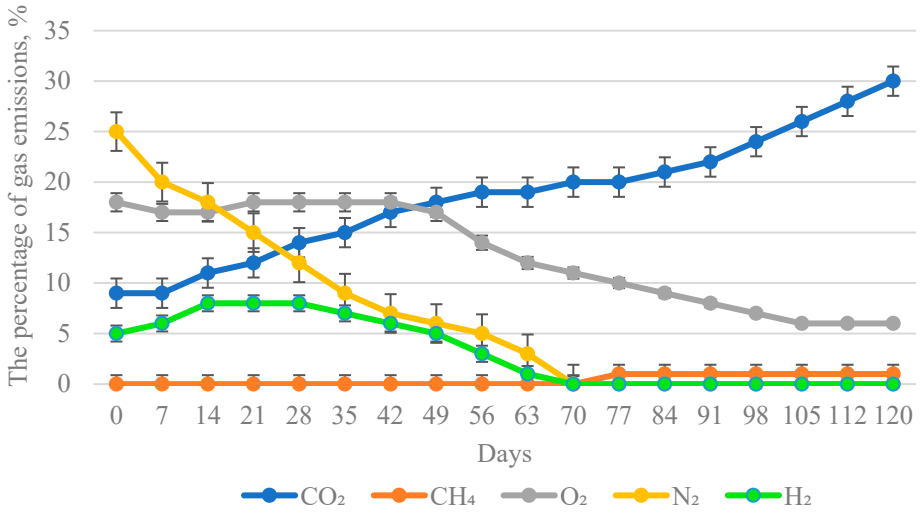


Figure 4. Gas emissions from the third landfill model.

Although aeration processes prevented the generation of methane and energy production from MSW waste, the advantages of this landfill model can also be observed. Such a model of biodegradation processes can be used to reduce gas emissions from already closed landfills. In this way, the filtrate volume is doubled, the emissions of carbon and nitrogen compounds are faster, and the formation of unwanted odors, usually during aerobic processes, is reduced. This aeration application can be applied to old landfills where waste stabilization and aeration are required to reduce pollutant emissions more quickly.

3.4. The Fourth Landfill Model

When applying an aerobic–anaerobic system, organic matter stabilization occurred significantly faster than during a purely aerobic system. Low air inflow, single water inflow, and leachate recirculation provided these conditions for the fourth landfill model. Analogously to the model of the third landfill, during the first 70 days of this study, N_2 and H_2 emissions decreased to 0%, respectively, with the first pollutant from 25% and the second from 5% (Figure 5).

Aeration processes significantly reduced COD (from 2.00 to 1.00 mg/L), TOC (from 0.20 to 0.10 mg/L), and BOD_5 (from 1.00 to 0.30 mg/L) of the filtrate because the oxidation of pollutants in aerobic conditions was significantly more intense. During the aerobic–anaerobic system, biogas production was improved due to the dominance of CO_2 (from 0 to 30%) and CH_4 (from 7 to 27%). During this process, the pH fluctuated around 7. Using the aerobic–anaerobic system model, it is possible to produce energy from MSW waste, as relatively large amounts of CO_2 and CH_4 were formed. Such a model could also reduce emissions of unwanted gasses from already closed landfills. In this way, with aerobic processes starting at the beginning of the experiment, the biodegradation processes could be accelerated to reduce pollutant emissions and the formation of unwanted odors. After analyzing the results of the third and fourth landfill models, this study's results are consistent with those obtained by other researchers [23,32–38]. According to them, aeration is one of the main ways to speed up the decomposition of municipal solid waste in landfills. Determining the proper aeration rate is critical to designing and operating a landfill aeration system. This makes it possible to increase biogas production (CO_2, CH_4) faster and reduce the concentration of different chemical substances (N_2, O_2, H_2).

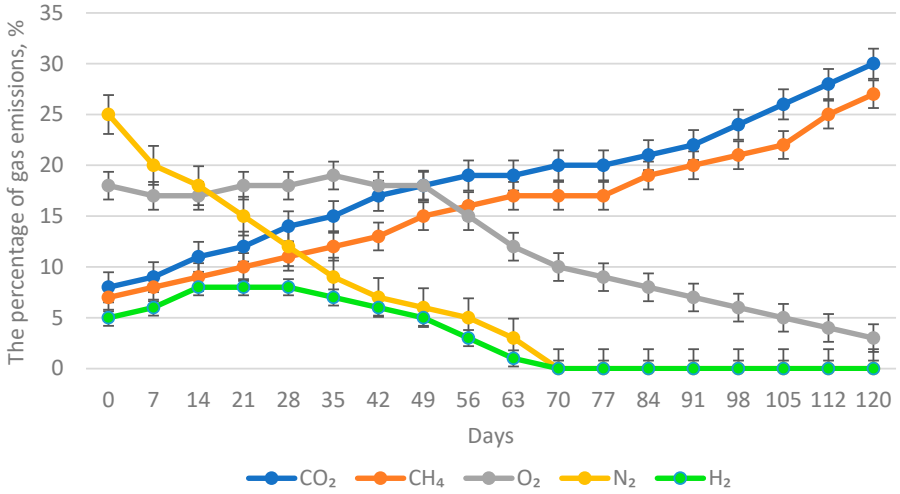


Figure 5. Gas emissions from the fourth landfill model.

3.5. The Fifth Landfill Model

The fifth aerobic–anaerobic and flushing bioreactor landfill had parameters different from the fourth. It had low air inflow, single water inflow, leachate recirculation, and a dose of probiotics, creating unique conditions for this landfill model. During the first 56 days of testing, aerobic nitrification–denitrification processes resulted in complete nitrogen removal (from 20% to 0%), and overall waste stabilization improved the biodegradation of the charge (see Figure 6). Biodegradation was enhanced by continuous leachate recirculation. At the same time, a decrease in H_2 from 2% to 1% was observed. After that, the activation of anaerobic processes decreased the amount of oxygen (from 14% to 2%) on days 56–120 of this study.

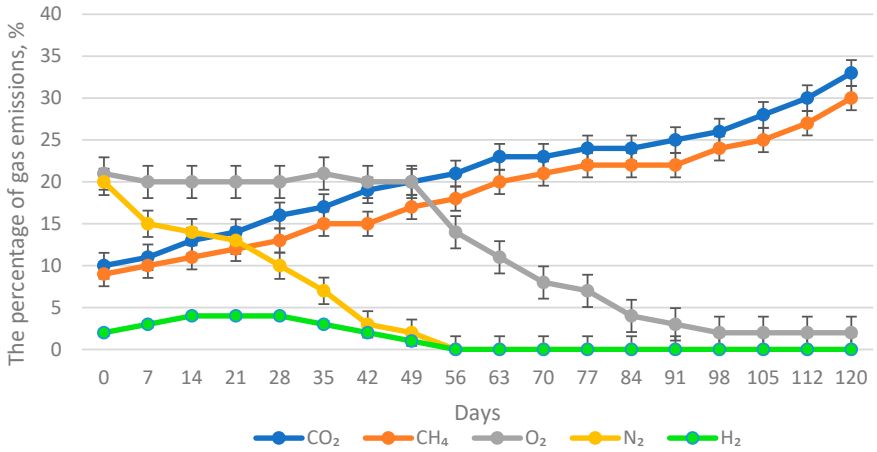


Figure 6. Gas emissions from the fifth landfill model.

During this period, there was a steady increase in the production of CH_4 and CO_2 and energy recovery. Accordingly, CO_2 increased from 10% to 33%, and CH_4 from 9% to 30%. Compared to the fourth landfill model, aerobic processes took place significantly more intensively (i.e., 14 days earlier). In addition, biogas production was accelerated by up to 10% by air flow supply, filtrate recirculation, and probiotic supply. A higher amount of filtrate and a dose of probiotics can maintain a higher and more stable methane concentration. Using probiotics in a fifth landfill model reduced odor and organic matter levels. During this stage, mesophilic bacteria and micromycetes oxidized the fermentation products of previous phases, such as other harmful gasses (hydrogen sulfide, sulfur mercaptan, light aromatic compounds). From an economic perspective, such technology is expensive, so it is only applied when traditional biodegradation processes cannot reduce pollutant emissions.

During this aerobic–anaerobic period, the aim was to maximize methane production by accelerating the acetogenesis phase and maintaining an optimal pH (7). Aeration processes significantly reduced COD (from 2.00 to 1.50 mg/L), TOC (from 0.20 to 0.15 mg/L), and BOD_5 (from 1.10 to 0.50 mg/L) of the filtrate because the oxidation of pollutants in aerobic conditions was significantly more intense. During this process, the pH prevailed between 7 and 7.2. The results of these studies and other researchers [22–26] confirmed that landfill aeration is an effective technique for the controlled and sustainable conversion of conventional anaerobic landfills into a biologically stabilized state associated with a significantly lower or near elimination of landfill gas emission potential.

3.6. The Comparison of the Analyzed Landfill Models

Discussing and comparing the general results of different analyzed landfill models was beneficial. The results of the research showed that the first and second models of landfills, using only anaerobic conditions, can be used for the treatment of MSW for the production of biogas (CH_4, CO_2), as up to 40–60% of it was released during the 120-experiment period. It can be noted that in the second landfill model, with the additional use of probiotics, biogas was released on average 10–15% more than in the first landfill model without probiotics. The third landfill model, which maintained only aerobic conditions, showed the lowest quantitative emissions of greenhouse gasses (for example, CH_4 was only at 1% during the 120-experiment period). This aeration program can be applied in old, already closed landfills, where the rapid stabilization and aeration of MSW is required to minimize pollutant emissions (N_2, etc.) and unwanted odors and shorten biodegradation processes. The results of the fourth and fifth landfill models, in which aerobic–anaerobic conditions were applied, showed that the developing nitrification–denitrification processes resulted in complete nitrogen removal (from 20% to 0%), and overall waste stabilization improved the biodegradation of the MSW. Later, relatively good (on average, 30%) results of biogas (CH_4, CO_2) emissions are achieved during anaerobic condition formation results. It can be noted that in the fifth landfill model, with the additional use of probiotics, biogas was released on average 10–15% more, and the processes took place on average 14 days more intensively than in the fourth landfill model without probiotics.

When analyzing the research results obtained from a practical point of view, the most important thing is to pay attention to biogas production in different landfill models. Analyzing all landfill models according to methane emissions, it was found that the most significant amount of this biogas was released in the first and second landfill models (Figure 7). As can be seen, up to days 63–70 of this research, the release of methane continuously increased (up to 60%), and then remained stable. Meanwhile, in the analyzed third, fourth, and fifth models of landfills, methane emissions ranged from 10 to 30%.

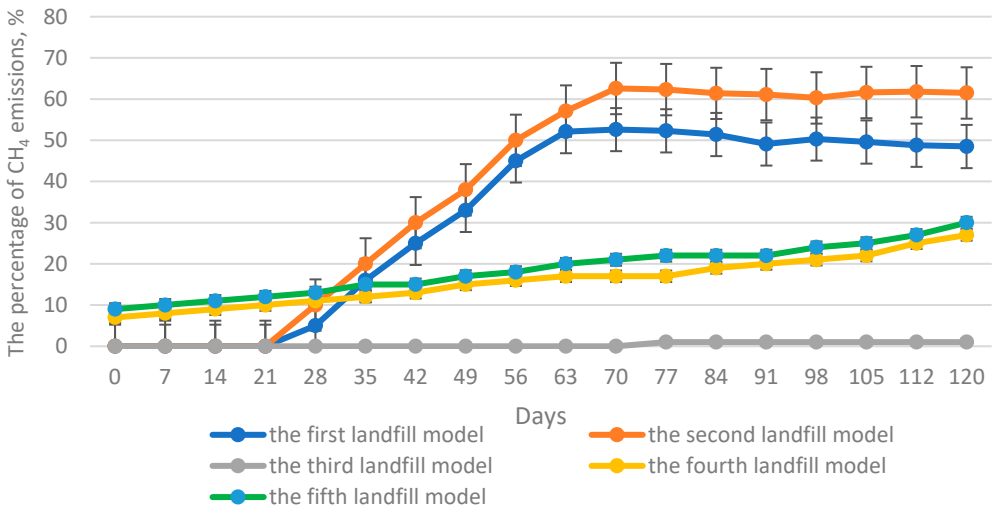


Figure 7. CH_4 emissions from the analyzed landfill models.

Similar research results were obtained for biogas, such as carbon dioxide, in the first and second landfill models (Figure 8). The highest emission peak of this gas (in the range of 60–70%) was recorded on the 35th day of research. Later, it decreased and fluctuated between 40 and 50%. Meanwhile, in the analyzed third, fourth, and fifth models of landfills, carbon dioxide emissions ranged from 10 to 30%.

The maintenance of anaerobic conditions determined better biogas (CH_4, CO_2) emissions in the first and second landfill models. Due to the incorporation of probiotics, biogas emissions in the second landfill model were, on average, 10% higher than in the first landfill model. These research trends are discussed in the first and second subsections of the Results and Discussion.

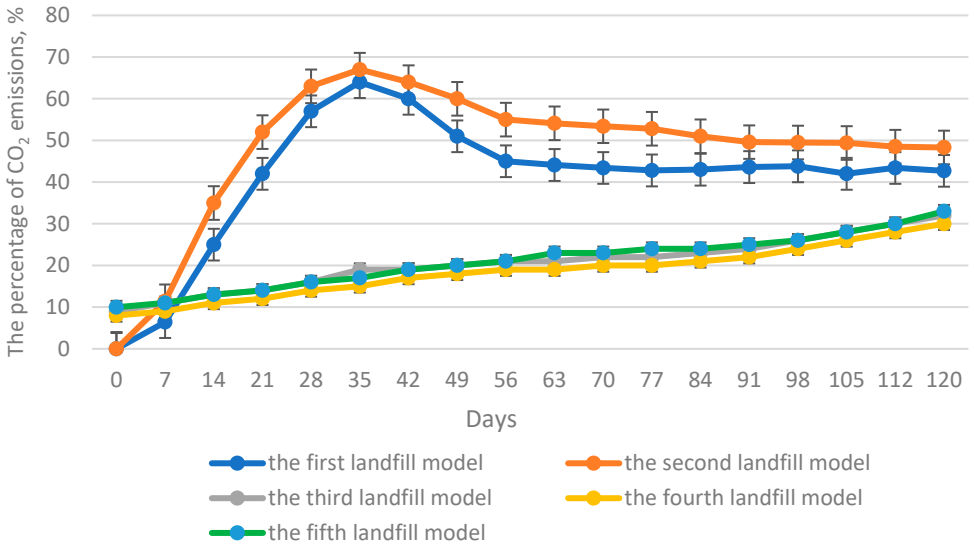


Figure 8. CO_2 emissions from the analyzed landfill models.

This study presented results that other researchers have analyzed [39–43]. Filtrate TOC, COD, and pH are the main parameters significantly influencing filtrate leaching. Analyses of leachate characterizations in five landfill models confirmed the results of the previously discussed studies (Table 3). COD, TOC, and BOD_5 were evaluated as indicators of leachate organic matter. In the first and second landfill models, COD, TOC, and BOD_5 concentrations increased due to anaerobic conditions and decreased due to the release of complex organic matter and simultaneous hydrolysis from waste to leachate. Accordingly, COD (from 5.00 to 2.00 mg/L), TOC (from 0.60 to 0.35 mg/L), and BOD_5 (from 3.00 to 0.50 mg/L) decreased. Meanwhile, in the third landfill model, in the filtrate with a pH in the range of 6.0–8.0, the indicators of organic matter in the filtrate were lower due to intensive aeration, which accelerated the biodegradation of organic matter. For example, COD (from 2.50 to 1.50 mg/L), TOC (from 0.35 to 0.20 mg/L), and BOD_5 (from 0.50 to 3.00 mg/L) decreased. Meanwhile, in the fourth and fifth aerobic–anaerobic landfill models, the least amount of organic matter was found compared to the previously examined landfill models due to the stabilization processes taking place in them. Accordingly, the results were as follows: COD (1.00–2.00 mg/L), TOC (0.10–0.20 mg/L), and BOD_5 (0.30–1.50 mg/L) decreased. Overall, these investigations can compare aerobic and anaerobic landfills for life-cycle assessment.

Table 3. The characterization (with a standard deviation (SD)) of the leachates in the analyzed landfill models.

Landfill Models	COD, mg/L	TOC, mg/L	BOD_5, mg/L	pH
1	3.00–5.00 (SD: ±0.04)	0.60–1.00 (SD: ±0.02)	1.00–3.00 (SD: ±0.05)	6.6–7.2 (SD: ±0.03)
2	2.00–3.00 (SD: ±0.05)	0.35–0.60 (SD: ±0.03)	0.50–2.00 (SD: ±0.04)	5.6–6.2 (SD: ±0.03)
3	1.50–2.50 (SD: ±0.02)	0.20–0.35 (SD: ±0.01)	0.50–1.50 (SD: ±0.03)	6.0–8.0 (SD: ±0.01)
4	1.00–2.00 (SD: ±0.05)	0.10–0.20 (SD: ±0.02)	0.30–1.00 (SD: ±0.04)	6.0–8.0 (SD: ±0.03)
5	1.50–2.00 (SD: ±0.03)	0.15–0.20 (SD: ±0.03)	0.50–1.10 (SD: ±0.05)	7.0–7.2 (SD: ±0.02)

4. Conclusions

By summarizing the experimental results of all landfill models for the further evaluation of processes, all models can be applied in actual practice depending on where they will be applied and what result is desired. That is, if it is necessary to produce more biogas, then it is recommended to use the first and second anaerobic landfill models, respectively. The methane production in the first one reached up to 53% and up to 63% in the second one.

If old landfills are to be remedied as quickly as possible, a third aerobic model should be applied, as the biodegradation processes significantly reduce pollutant emissions. For example, the exhaust gas composition decreased: O_2—from 18 to 6%, CO_2—from 30 to 9%, N_2—from 25 to 3%, and H_2—from 5 to 1%.

If treating MSW and obtaining more biogas is desired, only the fourth and fifth aerobic–anaerobic landfill models are the best because the biogas emissions have increased. Accordingly, in the fourth model, CO_2 emissions increased from 0 to 30%, and CH_4 emissions rose from 7% to 27%. Similarly, in the fifth model, CO_2 emissions increased from 10% to 33%, and CH_4 emissions rose from 9% to 30%.

These experimental studies showed that probiotics influenced the second and fifth landfill models and biogas production. For example, in the second model, methane emissions reached as high as 60–63% and remained stable until the end of the experiment.

To that end, it would be interesting to investigate probiotics' effectiveness shortly and clarify related questions: how quickly the amount of organic matter in landfill models changes, how quickly the number of microorganisms changes, how quickly the efficiency of pollutant cleaning or biogas production is achieved, etc.

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